

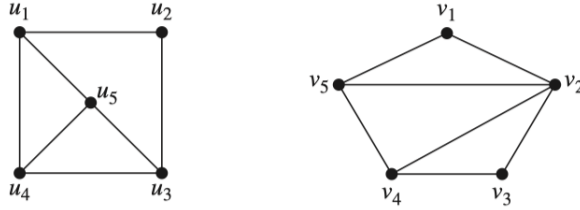
Exercise Sheet 12

Discrete Mathematics by Hongfei Fu, 2023.11.20

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1. ([R], Page 676-677, Exercise 36, 37, 42, 43(2pt), 44(2pt)) In Exercises 36, 37, 42, 43, 44 determine whether the given pair of graphs is isomorphic. Exhibit an isomorphism or provide a rigorous argument that none exists.

(36).



Answer Area:

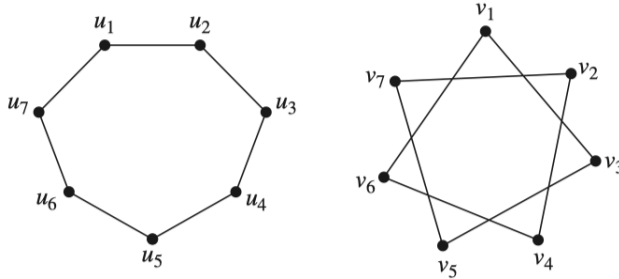
No. A proof is as follows:

$\because$ In the left graph, there isn't a vertex whose degree is 4.

But in the right graph, $\deg(v_2) = 4$.

$\therefore$ The two graphs are not isomorphic.

(37).



Answer Area:

Yes. There is a bijection $f: \mathbf{V}_1 \rightarrow \mathbf{V}_2$ such that $\forall u, v \in \mathbf{V}_1 (\{u, v\} \in \mathbf{E}_1 \leftrightarrow \{f(u), f(v)\} \in \mathbf{E}_2)$:

$$f(u_1) = v_1$$

$$f(u_2) = v_3$$

$$f(u_3) = v_5$$

$$f(u_4) = v_7$$

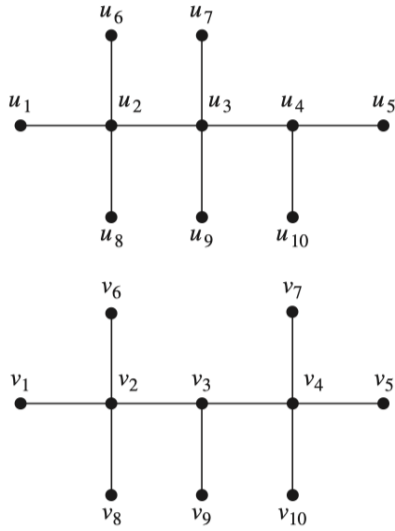
$$f(u_5) = v_2$$

$$f(u_6) = v_4$$

$$f(u_7) = v_6$$

Hence, the two graphs are isomorphic.

(42).



Answer Area:

No. A proof is as follows:

Assuming that there is a bijection $f: \mathbf{V}_1 \rightarrow \mathbf{V}_2$ such that $\forall u, v \in \mathbf{V}_1 (\{u, v\} \in \mathbf{E}_1 \leftrightarrow \{f(u), f(v)\} \in \mathbf{E}_2)$.

$\therefore$ If $f(u_i) = v_j$, then $\deg(u_i) = \deg(v_j)$

$\therefore f(u_5) = v_5$ or $f(u_5) = v_1$.

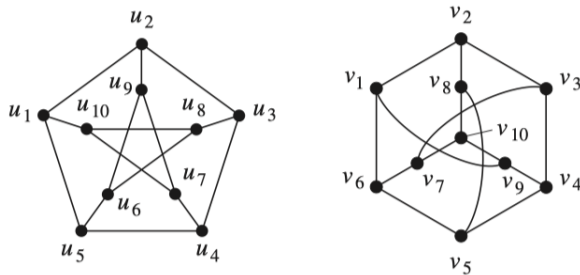
$\therefore \deg(v_2) = \deg(v_4) = 4 \neq 3 = \deg(u_4)$.

$\therefore f(u_2) \neq v_4$ and $f(u_2) \neq v_2$.

$\therefore f(u_5) \neq v_5$ and $f(u_5) \neq v_1$

Contradict. The two graphs are not isomorphic.

(43)(2pt).



Answer Area:

Yes. A bijection. There is a bijection $f: \mathbf{V}_1 \rightarrow \mathbf{V}_2$ such that $\forall u, v \in \mathbf{V}_1 (\{u, v\} \in \mathbf{E}_1 \leftrightarrow \{f(u), f(v)\} \in \mathbf{E}_2)$:

$f(u_1) = v_1$

$f(u_2) = v_2$

$f(u_3) = v_3$

$f(u_4) = v_7$

$f(u_5) = v_6$

$f(u_6) = v_5$

$f(u_7) = v_{10}$

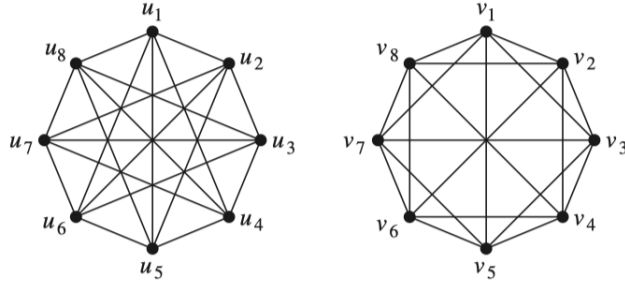
$f(u_8) = v_4$

$f(u_9) = v_8$

$f(u_{10}) = v_9$

Hence, the two graphs are isomorphic.

(44)(2pt).



Answer Area:

No. A proof is as below:

Let $\mathbf{G}_1 = (\mathbf{V}_1, \mathbf{E}_1)$, $\mathbf{G}_2 = (\mathbf{V}_2, \mathbf{E}_2)$ representing two graphs. Assuming that there is a bijection $f: \mathbf{V}_1 \rightarrow \mathbf{V}_2$ such that $\forall u, v \in \mathbf{V}_1 (\{u, v\} \in \mathbf{E}_1 \leftrightarrow \{f(u), f(v)\} \in \mathbf{E}_2)$.

According to the symmetry, we can define $f(u_1) = v_1$.

Also according to the symmetry, we can conclude that $f(u_5) = v_5$.

However, all the neighbours of u_5 are the neighbours of u_1 and some of the neighbours of v_5 aren't the neighbours of v_1 .

Therefore, contradict.

So there isn't such bijection. The two graphs are not isomorphic.